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A Deep Semi-Supervised Learning based Damage Identification of a Bridge Using Vehicle-Induced Dynamic Displacement and Multilayer Perceptron Model

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ABSTRACT

Structural health monitoring (SHM) of bridges is critical for maintaining infrastructure safety, but traditional methods like manual inspections and extensive sensor networks are often inefficient and costly. This study addresses these limitations by integrating a Multilayer Perceptron (MLP) neural network with the Dynamic Time Warping (DTW) algorithm, focusing on vehicle-induced displacement response data instead of traditional acceleration data. Utilizing vehicle-bridge interaction (VBI) modeling, the MLP is trained on undamaged bridge displacement data to predict future displacements, creating a baseline for healthy structural behavior. By comparing predicted and actual displacement data using DTW, a damage index (DI) is generated, where significant deviations indicate potential structural damage. The approach proves to be both cost-effective and robust, requiring fewer sensors and offering high accuracy due to the high signal-to-noise ratio of displacement data. The combination of MLP and DTW offers an efficient solution for real-time damage detection, paving the way for more affordable and practical SHM systems.

KEYWORDS: Structural Damage Identification, Bridge Structural Health Monitoring, Vehicle-Induced Dynamic Displacements, Displacement Response Measurement.

1. INTRODUCTION

Bridges play an integral role in national infrastructure, providing essential connectivity that promotes economic growth and supports the daily lives of millions by enabling efficient transportation across otherwise impassable areas such as rivers, valleys, and highways. Over time, these vital structures are subjected to a wide range of stressors, including continuous vehicle loads, environmental factors, and natural material degradation. These factors, together with the constant exposure to dynamic forces, make bridges particularly vulnerable to damage and deterioration, which can manifest as cracks, loose bolts, broken welds, corrosion, and other issues that compromise structural stability and safety [1]. For instance, the catastrophic failure of the I-35W River Bridge in Minnesota, which resulted in thirteen fatalities and significant economic costs,

highlights the profound impact that undetected structural damage can have on both human life and infrastructure resilience [2].

Traditionally, bridge inspections have relied on visual assessment methods. These methods, while straightforward, exhibit several limitations. They are inherently subjective, often varying based on the inspector's experience and judgment, and are also constrained by the interval between inspections, which can leave intermittent damage undetected for extended periods[3]. As a result, visual inspections, while helpful, may fall short in providing the level of reliability and responsiveness needed for timely detection and preventive maintenance. Addressing these challenges has led to growing interest in advanced structural health monitoring (SHM) systems, which enable continuous, data-driven evaluation of a bridge's condition. The goal of SHM is to improve detection accuracy, enhance response time, and ultimately support safer and more cost-effective infrastructure management.

In response to the need for more reliable and efficient monitoring solutions, this study proposes a novel approach to SHM, specifically aimed at detecting bridge damage early using a combination of Vehicle-Bridge Interaction (VBI) modeling, time series analysis, and machine learning techniques. The methodology influences numerical simulation of the VBI model, which integrates dynamic models of both the bridge and the vehicle subsystems, while accounting for critical variables such as vehicle speed, bridge dynamics, and road surface roughness [4]. By analyzing the vehicle-induced displacement responses of both undamaged and damaged bridge states, this research applies time series modeling to extract damage-sensitive features, which are further processed through machine learning algorithms. Specifically, a Multilayer Perceptron (MLP) neural network, used in conjunction with Dynamic Time Warping (DTW), enables accurate identification of structural damage through comparison of baseline and current bridge response data.

Overall, this research offers valuable insights into bridge health monitoring, presenting a practical approach to real-time damage detection. By advancing the technology available for SHM, this study contributes toward the development of safer, more resilient transportation networks, promoting extended service life for bridge infrastructure, optimizing maintenance strategies, and ultimately ensuring the safety and functionality of these critical assets. Through the integration of dynamic modeling, time series analysis, and machine learning, this study not only addresses existing challenges in bridge monitoring but also sets a foundation for future advancements in SHM practices that prioritize reliability, cost-effectiveness, and efficiency.

2. VEHICLE-BRIDGE INTERACTION SYSTEM

Vehicle Modeling: The dynamics of the vehicle has been modelled as a half car model as shown in **Figure 1** [5]. The equation of motions (EOM) of the vehicle dynamics are formulated in matrix form as in **Eq. (1)** [5].

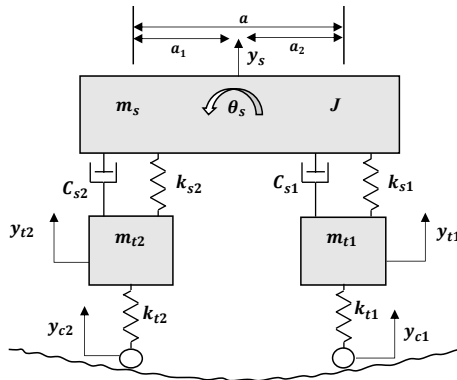


Figure 1: Half car vehicle model [5]

$$[M_v]\{\ddot{y}_v(t)\} + [C_v]\{\dot{y}_v(t)\} + [K_v]\{y_v(t)\} = \{F_v\} \quad (1)$$

Where, $[M_v]$ is the mass matrix, $[C_v]$ is damping matrix, $[K_v]$ is the stiffness matrix, $\{y_v(t)\}$ is the matrix of the degree of freedom of the vehicle, $\{F_v\}$ is the the exciting force of the vibration of the vehicle.

Bridge Modeling: The dynamics of the bridge has been modelled as a simply supported beam using the concepts of Finite Element Method (FEM) as shown in **Figure 2** [5]. The equation of motions (EOM) of the bridge dynamics are formulated in matrix form as in **Eq. (2)** [5].

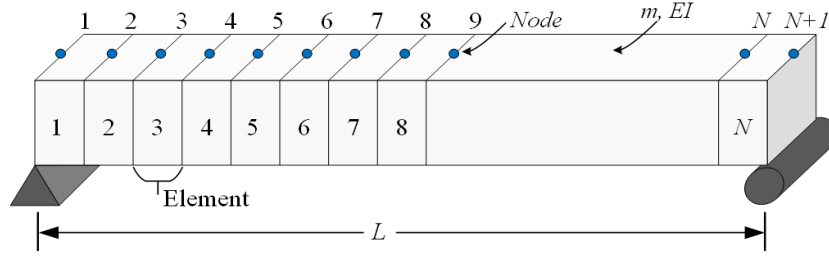


Figure 2: FE Model of the Bridge [5]

$$[M_b]\{\ddot{y}_b(t)\} + [C_b]\{\dot{y}_b(t)\} + [K_b]\{y_b(t)\} = [N_b]\{F_b(x, t)\}\delta(x - vt) \quad (2)$$

Where, $[M_b]$, $[C_b]$, $[K_b]$ are the mass, damping and stiffness matrices of the bridge, $F_b(x, t)$ denotes the coupled forces on the bridge and the vertical bridge displacement at nodal points at time t is represented by $\{y_b(t)\}$ and δ defines the Dirac function. $\{F_b(x, t)\}$ contains the interaction forces between the vehicle and the bridge.

Road Surface Roughness: The bridge pavement roughness is simulated theoretically, assuming vehicle wheels maintain contact and match pavement displacement. The resulting surface roughness, following the Class A-B profile per ISO 8608, is generated using **Eq. (3)** [6].

$$r(x) = \sum_{i=0}^N \sqrt{\Delta n} \cdot 2^k \cdot 10^{-3} \cdot \left(\frac{n_0}{i \cdot \Delta n} \right) \cdot \cos(2\pi \cdot i \cdot \Delta n \cdot x + \varphi_i) \quad (3)$$

Where, $r(x)$ is the elevation/roughness of the road profile; L represents the length of the bridge and B is the sampling interval; x represents the abscissa variable from 0 to L ; $\Delta n = 1/L$; $n_{max} = 1/B$; $N = n_{max}/\Delta n = L/B$; $n_0 = 0.1$ cycles/m; k denotes a constant value depending upon ISO road roughness classification [7], [8] which varies from 3 to 9, corresponding to the road roughness profiles from class A to class H. Also, φ_i represents random phase angle within the $0-2\pi$ range which follows a uniform probabilistic distribution.

Vehicle-Bridge Interaction Model: The coupled vehicle-bridge system is shown in **Figure 3** [5], where they interact at the tire contact points through interaction forces. The coupled equation of motion **Eq. (4)** [5] is formulated by combining **Eq. (1)** [5] and **Eq. (2)** [5].

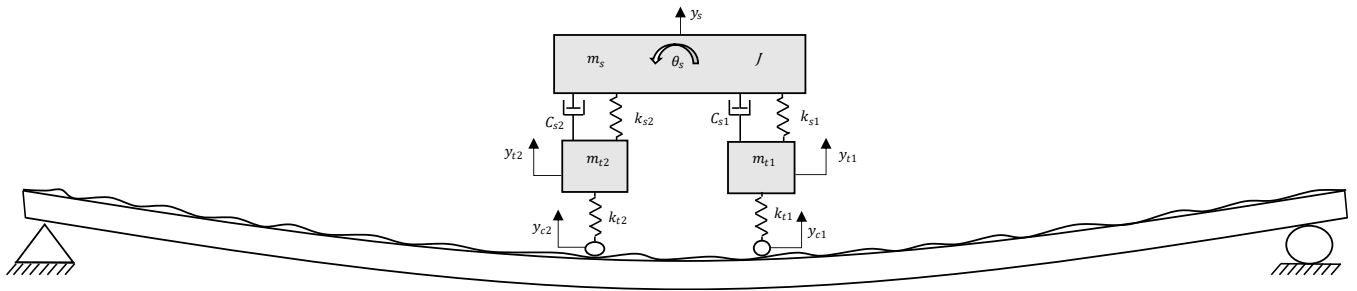


Figure 3: Coupled Vehicle-Bridge Vibration Model [5]

$$[M_g]\{\ddot{y}_g(t)\} + [C_g]\{\dot{y}_g(t)\} + [K_g]\{y_g(t)\} = \{F_g\} \quad (4)$$

Where $[M_g]$ and $[C_g]$ are the combine system mass and damping matrices respectively, $[K_g]$ is the coupled time-varying system stiffness matrix and $\{F_g\}$ is the system force vector.

3. METHODOLOGY

A Multilayer Perceptron (MLP) as in the domain of deep learning algorithm (DL) effectively handles complex, non-linear relationships between input data and outputs, making it ideal for modeling intricate systems like structural damage detection. MLPs can learn the dynamic behavior of healthy structures, such as bridges, and predict their behavior in both healthy and damaged states, facilitating the detection of anomalies. Their architecture, featuring multiple hidden layers and nonlinear activation functions, enables MLPs to capture intricate patterns in input data that simpler models may miss. The MLP consists of three main layers: the input layer, which processes raw data like vibration responses; the hidden layers, which transform inputs using weights, biases, and activation functions like ReLu to learn complex relationships [9], [10]; and the output layer, which predicts the vibration response and compares it to actual measurements to identify significant deviations indicating damage.

During training phase, the MLP model is trained on the vibration responses of a healthy bridge, using data collected from sensors as a baseline. Ten sets of vibration displacement responses, simulating a vehicle passing over the bridge, were generated with added noise to mimic real-world conditions. The data was normalized and augmented to create a robust training dataset, allowing the model to learn the structure's behavior in its healthy state by adjusting weights and biases to minimize the loss function. In the inspection phase, the trained MLP predicts the vibration response for both healthy and damaged states. The differences between the predicted and actual responses are captured as residuals, which serve as damage-sensitive features, indicating potential structural problems when discrepancies arise between the predicted healthy response and the actual damaged response.

The residual function for the undamaged state can be represented as:

$$e_u(t) = y_{pu}(t) - y_u(t) \quad (5)$$

The residual signal for the damaged state is derived as,

$$e_d(t) = y_{pd}(t) - y_d(t) \quad (6)$$

Where, $e_u(t)$ and $e_d(t)$ are the residual signals of the healthy and damaged state, respectively; $y_u(t)$ and $y_d(t)$ is measured vibration data of the undamaged state and damaged state, respectively; y_{pu} is predicted time series data obtained by modeling response of the undamaged state and $y_{pd}(t)$ is predicted time series data obtained by modeling response of the damaged state based on the model parameters of the undamaged state.

To detect structural damage, the MLP model's residuals from healthy and damaged states are compared using Dynamic Time Warping (DTW) to measure dissimilarity between the residual time series. DTW is a method that identifies the best alignment between two time series, allowing for nonlinear adjustments such as stretching or compressing the time axis of one series relative to the other. This alignment can be used to find the distance between the two time series data [11]. **Eq. (7)** $DTW(e_{u1}, e_d)$ represents the value between residuals in the undamaged (e_{u1}) and damaged (e_d) states, while in **Eq. (8)** $DTW(e_{u1}, e_{u2})$ indicates the value between two residuals from undamaged states. If damage is present, the residuals will significantly differ, leading to a higher DTW value for $DTW(e_{u1}, e_d)$ compared to $DTW(e_{u1}, e_{u2})$. This increased DTW value serves as the Damage Index (DI), reflecting a change in the structure's behavior due to damage. Analyzing the distribution of DTW values across 10 sets of vibration data helps determine the presence and extent of damage.

$$DI = DTW(e_{u1}, e_d) = d_{\text{optimal}}(e_{u1}, e_d) = \min\{d_{\text{optimal}}\} \quad (7)$$

$$DI = DTW(e_{u1}, e_{u2}) = d_{\text{optimal}}(e_{u1}, e_{u2}) = \min\{d_{\text{optimal}}\} \quad (8)$$

4. Numerical Investigation

Half car model is used to numerically generate the displacement response. The bridge parameters are provided in the **Table 1 [5]**, and the half car model parameters are taken from Rana et al. (2022) [5].

Table 1. Parameters of bridge [5]

Bridge Span, L (m)	Mass per unit length, m (kg/m)	Flexural rigidity, EI (Nm ²)	Modal Damping Ratio, ξ (%)	ISO road roughness classification, k
50	6818.5	6.9618×10^{10}	5	3

Displacement sensor is positioned three-quarters of the span from the left, as shown in the **Figure 4**. When the vehicle enters the bridge, the sensor will begin capturing the displacement response until the vehicle exits the bridge.

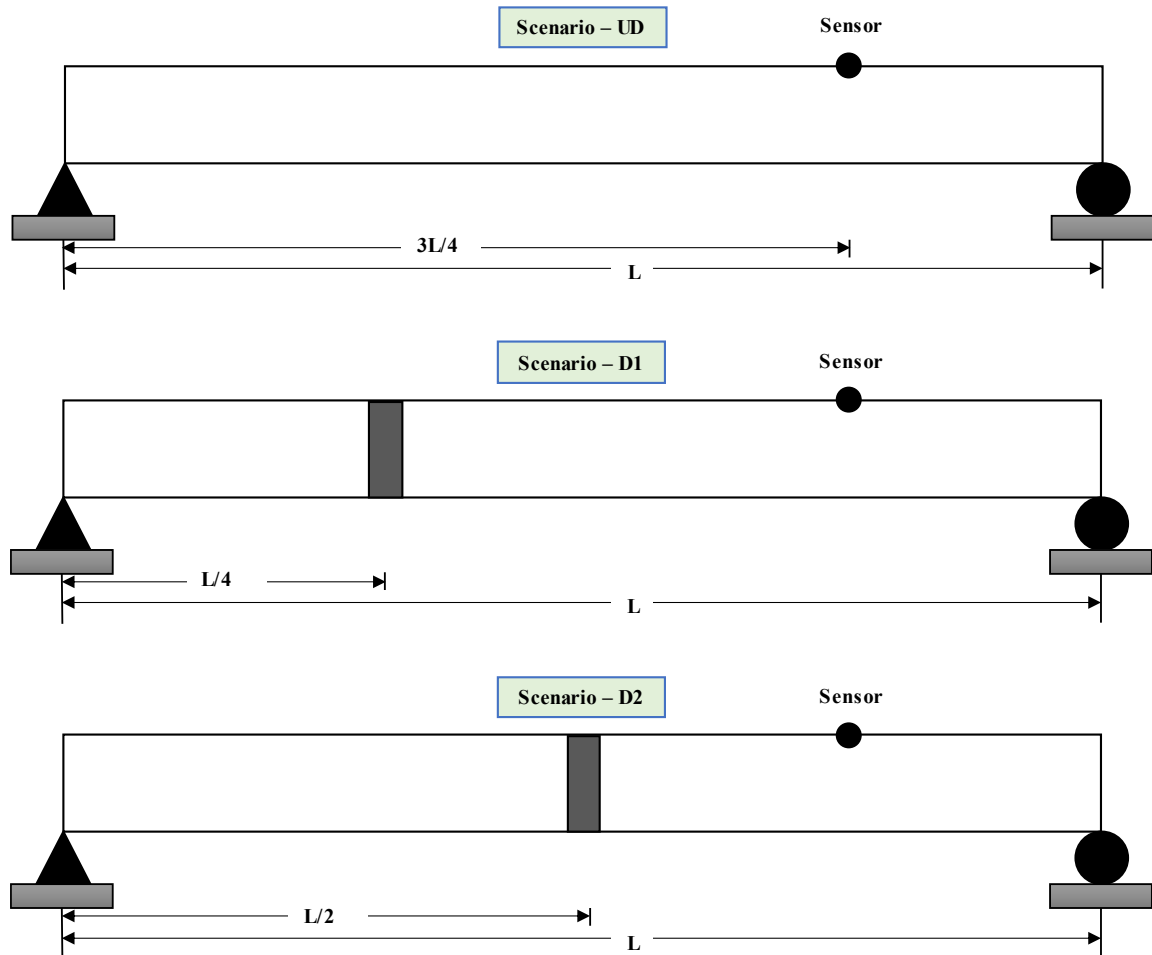


Figure 4. Sensor Placement and Damage Scenarios in Bridge

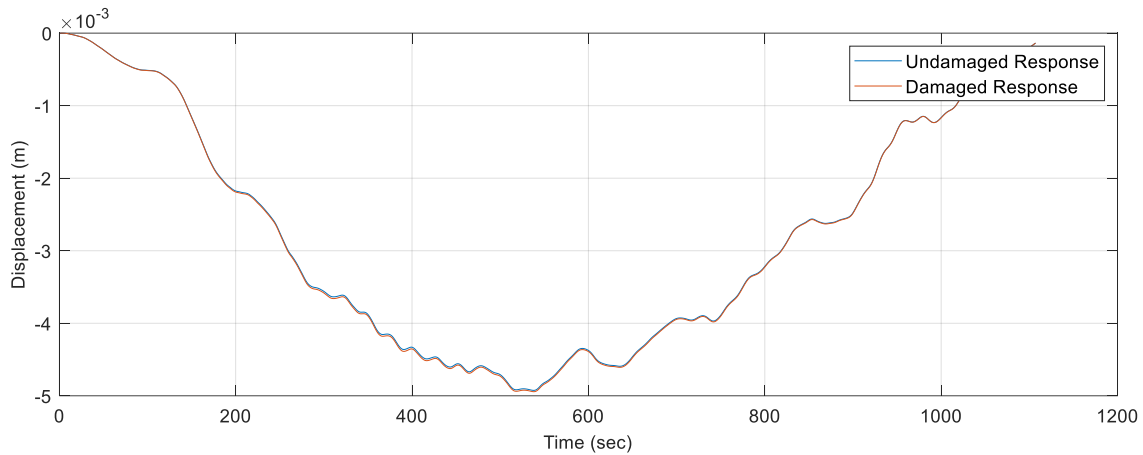
In the bridge model, damage is simulated by reducing the stiffness of a specific bridge element by 40%. This study considers two damage scenarios as shown in **Figure 4**, both involving single damage. In the first scenario, the damage occurs at one-fourth of the span from the left, which is the farthest position from the sensor. In the second scenario, the damage occurs at the middle of the bridge, which is closer to the sensor than the first damage position.

Table 2. Damage Scenarios

Type	Scenario	Damage Location	Damage severity, α (%)
Undamaged	UD	-	-
Single Damage	D1	at one-fourth	40
	D2	at half	40

Vehicle-induced vibration responses from the reference structure were collected as the baseline (healthy) state using displacement sensors. The response data was numerically prepared, and noise was added to simulate real-life data pollution. After collection, the data was normalized and augmented to create 10 sets of vibration displacement responses, each representing a vehicle passing over the bridge. This approach provided a robust dataset for training the MLP model.

The Vehicle-Bridge Interaction coupled equations as shown in **Eq. (4)** [5] were solved using the Newmark beta method to determine the bridge's dynamic displacement response, with data collected at designated sensor locations. Raw displacement data, augmented with 2% noise, was normalized and used for training the MLP model. A typical displacement data is shown in **Figure 5**. The MLP model consisted of five fully connected layers (as shown in **Table 3**), using ReLu activation for non-linearity and dropout regularization with a 0.1 probability to prevent overfitting. During training, data from the undamaged bridge, with the vehicle moving at 36 km/h, were collected at 200 Hz, resulting in 1,107 displacement response points per sensor. To detect anomalies, ten additional datasets with 2% noise were generated, creating 11,070 data points for training.

**Figure 5: Typical Damage – Undamaged Displacement Response**

Each dataset was segmented into sequences, yielding $(1107 - 32(\text{sequence length}))$ sequences for model training. A sequence length of 32, 30 training epochs, and a mini-batch size of 128 were selected. The MLP model parameters were tuned through trial and error, minimizing RMSE and producing an R^2 value closest to 1. The final MLP regression model parameters are detailed in **Table 3**. After training, the model was tested on a separate dataset to evaluate its performance.

Table 3: MLP Regression Architecture

Name	Operation	Dimension
"sequence input"	Sequence Input	Sequence input with 32 dimensions
"fc"	Fully Connected	128 fully connected layers
"relu"	reluLayer	
"fc"	Fully Connected	64 fully connected layers
"relu"	reluLayer	
"fc"	Fully Connected	32 fully connected layers
"relu"	reluLayer	
"fc"	Fully Connected	16 fully connected layers
"relu"	reluLayer	
"fc"	Fully Connected	1 fully connected layers
"regression layer"	Regression Output	

5. RESULTS AND DISCUSSION

After obtaining the baseline model from the training phase, the model's performance is evaluated in the testing phase for both the undamaged and damaged bridge conditions. Both cases show an R^2 value closer to 1, indicating higher model reliability, and an RMSE value closer to 0, which suggests accurate predictions for each vehicle pass. The damage index is then computed using residuals and used to differentiate between undamaged and damaged cases.

Figure 6 illustrates the damage index of the undamaged case (UD) and damage scenario (D1) generated from the trained MLP model. **Figure 6** shows that the trained MLP model yielded a greater DI for the damaged case than the undamaged case. As a result, the undamaged (UD) and damaged (D1) case are easily separable.

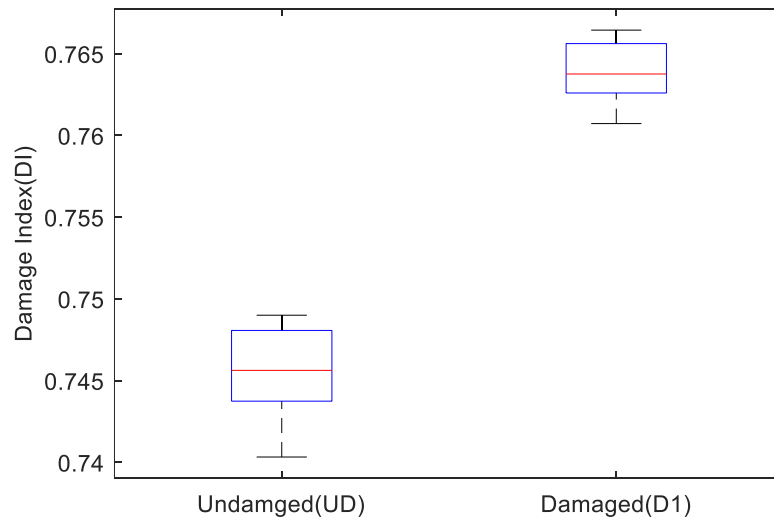
**Figure 6. Damage Index of UD and D1**

Figure 7 illustrates the damage index of the undamaged (UD) and damaged scenario (D2) obtained from the trained MLP model. **Figure 7** shows that the trained MLP model yielded a greater DI for the damaged case than the undamaged case. As a result, the undamaged case (UD) and the damaged case (D2) are readily distinguishable. Since the sensor is closer to D2 than D1, D2 has a greater damage index than D1.

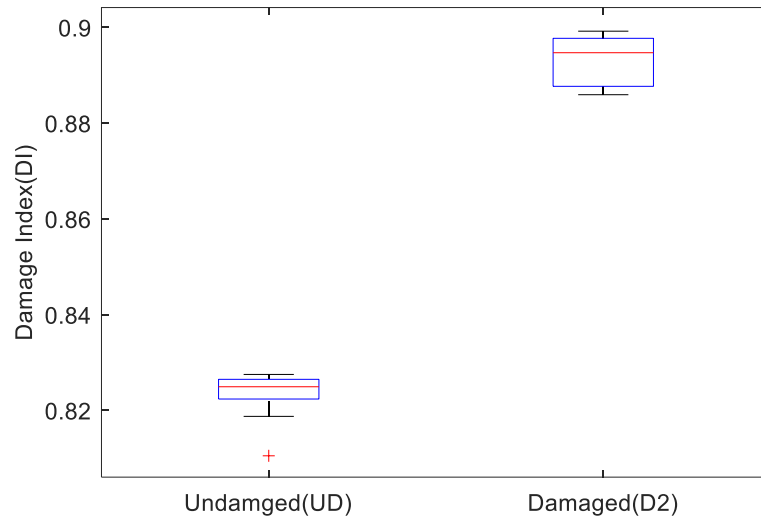


Figure 7. Damage Index of UD and D2

6. CONCLUSION

This paper introduces a method for detecting bridge damage using a Multi-Layer Perceptron (MLP) model. The approach utilizes displacement response data from a simply supported girder bridge under a two-axle moving vehicle load, tested in two different damage scenarios. By analyzing displacement data, the MLP model is able to extract damage-sensitive features effectively. Key conclusions from this study include:

- The proposed MLP model shows a low damage index for undamaged bridge data and a high damage index for damaged data, reliably identifying damage regardless of its location on the bridge.
- This method requires only displacement data from undamaged bridges, making it straightforward and practical for real-world applications.
- Damage closer to the sensor produces a higher damage index than damage located further away.

Although this method was validated through numerical simulations, experimental testing is still required. This study used only one sensor and one damage level, which future research could expand upon to enhance accuracy. Overall, this study offers a valuable contribution to bridge monitoring by providing an efficient and practical tool for detecting damage, supporting real-world bridge maintenance and safety.

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REPONSE TO REVIEWERS' COMMENTS: CLARIFICATION AND REVISIONS

Reviewers' Comments	Author Responses/Actions Taken
The author reconstructed the numerical model from reference [5] for the bridge's Structural Health Monitoring (SHM)	Thank you for this observation.
Several reference citations [5] are accessible	Thank you for your comment. If the concern is redundancy or accessibility, please let us know so we can address it.
Only one curve with two legends is shown in Figure 5	The issue arises because the two curves overlap, indicating the similarity of Damage and Undamaged Displacement Response data.
Equation numbering is improperly handled on page 4	The equation numbering of all pages (including page 4) has been corrected for consistency and accuracy.
In page 5, Figure no. is missing in text	The missing reference in the text on page 5 is not a figure but a table, which has been added.
Furthermore, the article's comprehension could not be adequately addressed	We reviewed the manuscript, made minor edits for clarity, and improved the structure. Please let us know if specific sections need further elaboration.